Shooter hardware findings

Four POLLEN fired in a burst land as roughly two long and two short. Software changes are being tested separately. These are the mechanical causes worth checking, in the order I would look at them. Numbers marked as estimates come from the code and published specs, not from measurements on the robot.

The ball

- The manual describes POLLEN as approximately 2.8 in (7.1 cm) polyethylene balls, and says they are not perfectly spherical and may vary in size.
- Vendor listings give 2.8 in ± 0.1 in and 0.055 lb. That is 71.1 mm ± 2.5 mm and about 25 g.
- The ball is hard and light. Almost all the give in the shot has to come from the 30A wheel and the silicone roller.

1. Compression is smaller than the ball tolerance

With a 1 mm squeeze on a nominal ball, the vendor tolerance alone moves the squeeze from negative to 3.5 mm. **A small ball may barely touch.** Grip then changes ball to ball, so exit speed changes even at identical wheel speed.

Ball diameter	Current gap (1 mm nominal)	4 mm nominal
68.6 mm (small)	-1.5 mm	1.5 mm
71.1 mm (nominal)	1.0 mm	4.0 mm
73.7 mm (large)	3.5 mm	6.5 mm

- Measure ten of our own balls with calipers first, across two axes each. The real spread sets the number.
- Aim for a nominal squeeze about 1.5 mm larger than the measured spread, so the smallest ball is still gripped. On vendor tolerance that is about 4 mm.
- More squeeze costs more wheel speed per shot. Test 2, 3 and 4 mm and keep the tightest grouping.

2. The fixed side rails are the other half of the grip

With the lower rollers removed, the ball rides on the stationary side rails as intended and does not touch the standoffs. Testing so far has been with the hood down only. This is a lower priority than finding 1, but it makes finding 1 worse.

- Against rails that do not move, the wheel rolls the ball along at about half the wheel's surface speed. That only holds if the ball rolls on the rails. A hard polyethylene ball on a hard rail with 1 mm of squeeze can slide instead, by a different amount each shot.
- The ball sits between two rails, so a smaller ball drops deeper between them. Ball size variation therefore changes the squeeze by a little more than the diameter change itself.

- A grippy or slightly soft surface on the rails would make the ball roll reliably and add give on that side.
- The driven top roller then takes the ball from about half speed to full speed in a short contact. Check that roller diameter × belt ratio equals the 72 mm wheel diameter, so the ball is not scrubbed between two surface speeds.

3. The mechanism still has a lot of drag

The controller needs about 0.57 power to hold 1530 RPM. A bare 6000 RPM motor turning at a quarter of its free speed should need about half of that.

Estimate: roughly 30% of stall torque, about 3 A and 7 W, is going into friction continuously.

- Coast-down test: spin to 1530 RPM, cut power, time how long the wheel takes to stop. With this much stored energy it should run for many seconds. One or two seconds confirms the drag.
- Likely places: belt tension, the 2:1 roller stage, bearing side load from the plates, and the washers against the bearings.
- Less drag gives the motor more torque left over to recover between balls.

4. Stored energy against energy per shot

Estimate, assuming 72 mm flywheel discs and the roller turning at twice wheel speed: about 5 to 7 J stored at 1530 RPM. Each ball leaves with about 0.4 J, plus losses.

- That is roughly 3 to 5% of wheel speed lost per ball if the motor has not recovered, which is 7 to 10% of range per ball.
- Four balls in quick succession stack these losses. This matches the long-then-short pattern.
- Stored energy goes with speed squared. A flywheel on the faster roller shaft stores four times what the same flywheel stores on the main shaft, as long as its belt does not slip.

Tests that separate the causes

1. **Single shots:** fire four balls with two seconds between each. If they still scatter, it is grip (finding 1, then 2). If they group, it is recovery (findings 3 and 4, plus software).
2. **Sorted balls:** fire the three smallest and three largest measured balls singly. A clear difference confirms finding 1.
3. **Coast-down:** as in finding 3.
4. **Burst log:** the shooter logging OpModes write wheel RPM and motor power to a CSV. The dip per ball shows directly how much speed each shot costs.

Sources

- BIOBUZZ Competition Manual, Section 9 (Arena): ftc-resources.firstinspires.org/ftc/game/manual-09
- Studica, BIOBUZZ POLLEN scoring element (tolerance and weight): studica.ca/en/biobuzz-pollen-game-preview-pack